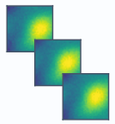
(a) VCEA: Validation-Calibrated Aggregation

1) Expert Pool

Input tensors (multi-modal)



…

L **historical frames**

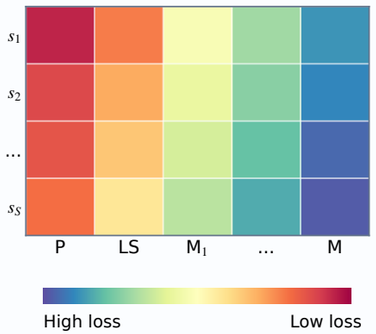
- **Persistence (P)**
- **Linear-Seasonal (LS)**
- **M1/C2 neural expert**
- **C3 opt neural expert**

2) Validation-only calibration

Validation MSE per tile/horizon:

$$L^{\text{val}}_{s,h,m} = \text{MSE}_{\text{val}}\left(Y, \hat{Y}^{(m)}\right)$$

Example: horizon h



3) Persistence-normalized inverse-loss weights

For each (s,h)**:**

$$a_{s,h,m} = \frac{1}{L^{\text{val}}_{s,h,m} + \varepsilon}$$

(normalized by persistence)

Floor and normalize:

$$w_{s,h,m} = \frac{\max(a_{s,h,m}, \eta)}{\sum_j \max(a_{s,h,j}, \eta)}$$

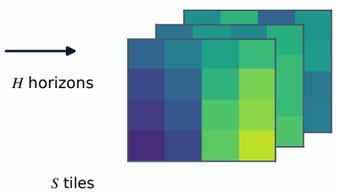
floor η

$\sum_m w_{s,h,m} = 1$

4) Tile × horizon static weights

Stored weight matrix

$$W \in \mathbb{R}^{S \times H \times M}$$

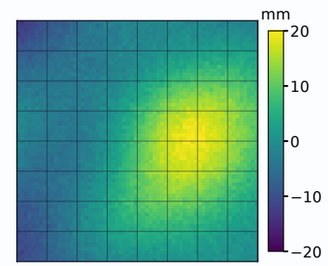


Frozen after validation (no test/audit updates).

5) VCEA prediction

For test time (s,h):

$$\hat{Y}^{\text{VCEA}}_{s,h} = \sum_{m=1}^{M} w_{s,h,m} \hat{Y}^{(m)}_{s,h}$$



VCEA nowcast: 64×64 EGMS displacement tile

(b) SS-VCEA: Stress-Safe Fallback Wrapper

1) Candidate predictions

- **VCEA prediction** $\hat{Y}^{\text{VCEA}}_{s,h}$
- **COEA prediction** $\hat{Y}^{\text{COEA}}_{s,h}$
- **Persistence prediction** $\hat{Y}^{P}_{s,h}$

2) Three-component risk score

For each (s,h):

Disagreement: $d_{s,h}$: VCEA--COEA

Complexity: $c_{s,h}$: COEA--Persistence

Volatility: $v_{s,h}$: persistence change

$$r_{s,h} = \frac{d_{s,h} + c_{s,h} + v_{s,h}}{3}$$

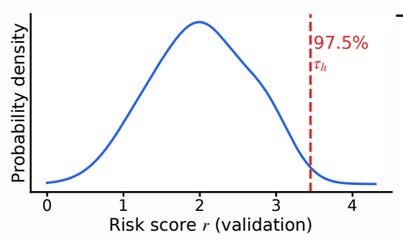
Higher r = higher risk (less trustworthy VCEA)

3) Aggregate risk

4) Validation threshold

For each horizon h:

$$\tau_h = q_{97.5}\left(r^{\text{val}}_{s,h}\right)$$



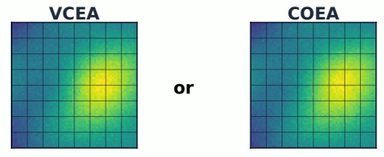
5) Safety-aware decision

If $r_{s,h} > \tau_h$:

Use COEA (safer)

Else:

Use VCEA (more accurate)



Legend

- Accuracy path (VCEA)
- Stress-safe wrapper (SS-VCEA)
- COEA fallback
- Data / tensors

Key Notation

- s : tile index $(1,\ldots,S)$
- h : forecast horizon $(1,\ldots,H)$
- m : expert index $(1,\ldots,M)$
- Y : ground truth
- $\hat{Y}^{(m)}$: prediction from expert m
- MSE : mean squared error

Protocol Guarantees

- ✓ Validation-only weights and thresholds.
- ✓ No test or external-audit calibration.
- ✓ External audit remains frozen.

- Data / computation flow
- Calibration (validation only)